

We went over conversions from base $n \rightarrow$ decimal.

What about decimal $\rightarrow$ base n ?

Equivalent numbers - numbers written in different bases that have the same decimal equivalent value.

Ex: $(0011)_8 = 9$
 $(1001)_2 = 9$ } The binary number and octal number are equivalent because they both convert to $(9)_{10}$

When converting from decimal $\rightarrow$ base n , we do it in two parts

1) The integer part (division). These are the digits left of the radix.

Ex: $(41)_{10} \rightarrow (?)_2$

$$\begin{array}{rcl} 41 \div 2 & = & 20 \text{ R: } 1 \\ 20 \div 2 & = & 10 \text{ R: } 0 \\ 10 \div 2 & = & 5 \text{ R: } 0 \\ 5 \div 2 & = & 2 \text{ R: } 1 \\ 2 \div 2 & = & 1 \text{ R: } 0 \\ 1 \div 2 & = & 0 \text{ R: } 1 \end{array}$$

- Divide the decimal number by the base, n

- Write down the remainder

- Divide the quotient again by n

- Keep going until you reach 0

- Read remainders from bottom to top

$(41)_{10} \rightarrow (101001)_2$

Ex: $(51)_{10} \rightarrow (?)_3$

$$\begin{array}{rcl} 51 \div 3 & = & 17 \text{ R: } 0 \\ 17 \div 3 & = & 5 \text{ R: } 2 \\ 5 \div 3 & = & 1 \text{ R: } 2 \\ 1 \div 3 & = & 0 \text{ R: } 1 \end{array}$$

$(51)_{10} \rightarrow (1220)_3$

2) The Fraction part (multiplication). These are the digits right of the radix

Ex: $(0.6875)_{10} \rightarrow (0.?)_2$

$$0.6875 \times 2 = \underline{1}.375$$

$$0.375 \times 2 = \underline{0}.75$$

$$0.75 \times 2 = \underline{1}.50$$

$$0.50 \times 2 = \underline{1}.00 \quad \downarrow$$

• Multiply the fraction number by base n

• Write down the integer part of the answer

• Take the new fraction part and multiply

• Keep going until you reach 0

• Read integers from top to bottom.

$$(0.6875)_{10} \rightarrow (0.1011)_2$$

For some fraction conversions, we may never exactly hit 0.

Ex: $(0.513)_{10} \rightarrow (0.?)_8$

$$0.513 \times 8 = \underline{4}.104$$

$$0.104 \times 8 = \underline{0}.832$$

$$0.832 \times 8 = \underline{6}.656$$

$$0.656 \times 8 = \underline{5}.248$$

$$0.248 \times 8 = \underline{1}.984 \quad \downarrow$$

$\vdots \quad \vdots \quad \vdots$

We can simply round to 4 decimal places

$$(0.513)_{10} \approx (0.4065\dots)_8$$

To convert a decimal number that has both integer and fraction parts, do each separately, then join with a radix point

Ex: $(41.6875)_{10} \rightarrow (?)_2$

$$(41)_{10} = (101001)_2$$

$$(0.6875)_{10} = (0.1011)_2$$

$$(41.6875)_{10} \rightarrow (101001.1011)_2$$

Hexadecimal = base 16 number system } hexadecimal, octal, base 4, and binary
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ¹⁰A, ¹¹B, ¹²C, ¹³D, ¹⁴E, ¹⁵F] all share relationships

$\log_2 4 = 2$ } Each base 4 digit is equal
to 2 binary bits

$\log_2 8 = 3$ } Each octal digit is equal to
3 binary bits

$\log_2 16 = 4$ } Each hexadecimal digit is equal
to 4 binary bits

Binary → Octal Conversion

- For integers, move from right to left, splitting into groups of 3
- For fractions, move from left to right, splitting into groups of 3
- Convert each binary term.

Ex: $(10110001101011.111100000110)_2 \rightarrow (?)_8$

010 110 001 101 011 . 111 100 000 110

2 6 1 5 3 . 7 4 0 6

$(26153.7406)_8$

← If a group is missing bits, you
can add leading \ trailing 0s.

Octal → Binary Conversion:

- Split and replace each digit with its 3-bit binary equivalent

Ex: $(673.124)_8 \rightarrow (?)_2$

6 7 3 . 1 2 4 = $(110111011.001010100)_2$

110 111 011 . 001 010 100

Binary $\rightarrow$ hexadecimal:

- For integers, move from right to left, splitting into groups of 4
- For fractions, move from left to right, splitting into groups of 4
- Convert each binary term. For any digit greater than 9, use its letter equivalent.

Ex: $(10110001101011.11110010)_2 \rightarrow (?)_{16}$

0010 1100 0110 1011.1111 0010

2 12 6 11 . 15 2

2 C 6 B . F 2

$(2C6B.F2)_{16}$

Hexadecimal $\rightarrow$ Binary:

- Convert letters to their numerical value
- Split and replace each digit with its 4-bit binary equivalent

Ex: $(306.D)_{16} \rightarrow (?)_2$

3 0 6 . D

3 0 6 . 13

0011 0000 0110 . 1101 = $(001100000110.1101)_2$

Converting Octal $\rightarrow$ hexadecimal or hexadecimal $\rightarrow$ octal will require two steps:

- 1) convert to binary from starting base
- 2) convert to desired base from binary